ML in Finance: Principal Component Analysis of Global Markets

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1 Introduction

Principal Component Analysis (PCA) is a foundational dimensionality reduction technique used in quantitative finance to identify latent drivers of asset returns[cite: 5]. Whether analyzing the term structure of interest rates or the sector rotation of the S&P 500, PCA allows us to distill hundreds of correlated variables into a few interpretable factors[cite: 6].

2 Mathematical Foundation

2.1 Data Representation and Stationarity

To ensure the data is stationary, we transform the raw prices or yields[cite: 11]:

- **Interest Rates:** We utilize *first differences* ($\Delta y_t = y_t - y_{t-1}$) because yields are already expressed in percentages[cite: 12].
- **Equities:** We utilize *percentage changes* or returns ($r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$) to normalize for absolute price levels[cite: 13].

2.2 The Covariance Matrix

PCA seeks to find the directions of maximum variance[cite: 15]. Let X be the $T \times n$ matrix of centered changes. The sample covariance matrix Σ is[cite: 15]:

$$\Sigma = \frac{1}{T-1} X^T X$$

2.3 Eigen-Decomposition

We diagonalize Σ to find the principal components[cite: 20]:

$$\Sigma = Q \Lambda Q^T$$

where Λ contains the eigenvalues and $Q = [v_1, v_2, v_3]$ contains the eigenvectors[cite: 22].

3 Cross-Asset Interpretation

The first three principal components often explain over 95% of market movement[cite: 25].

Component	**Fixed Income Interpretation**	**Equity Interpretation**
PC1	Level (Parallel Shift)	Market Factor (Beta)
PC2	Slope (Twist)	Sector/Style Rotation
PC3	Curvature (Butterfly)	Specific/Thematic Clusters

Table 1: Comparison of latent factors[cite: 27].

4 Historical Case Studies

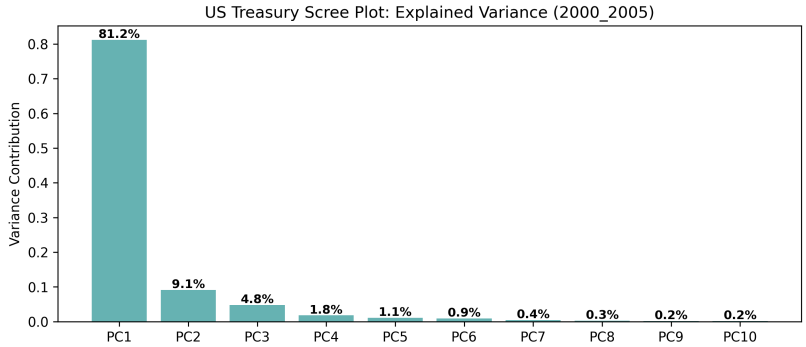
4.1 The 2008 Global Financial Crisis (2005–2010)

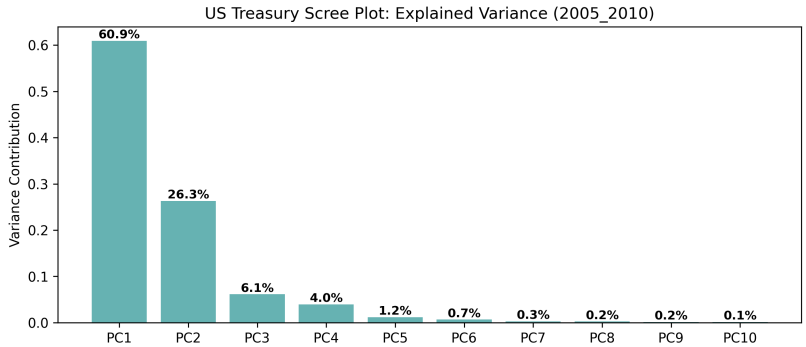
During this period, the Financials (XLF) sector showed a disproportionately high loading in PC1 and PC2[cite: 31]. In Fixed Income, the "Slope" factor became highly volatile as the Fed cut rates[cite: 32].

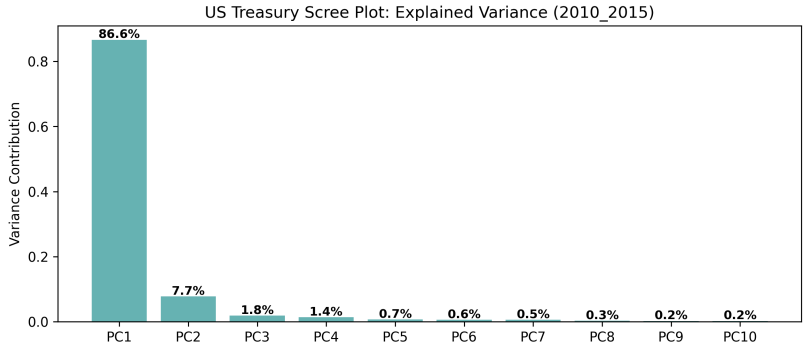
4.2 The Pandemic and Inflation Era (2020–2026)

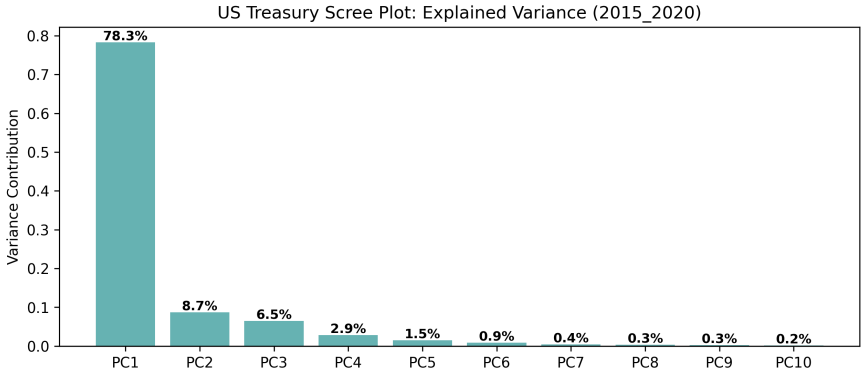
Technology (XLK) has dominated the Market Factor[cite: 34]. In the Treasury market, the 2022–2023 hiking cycle saw the "Level" factor regain nearly 90% of explained variance[cite: 35].

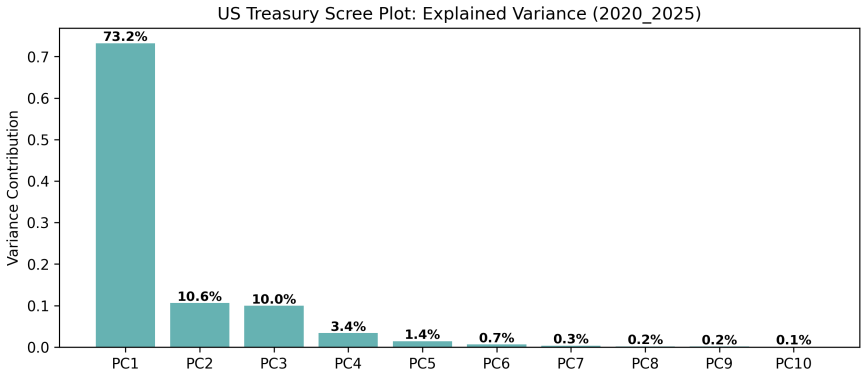
A Interest Rate: Scree Plots (2000–2025)



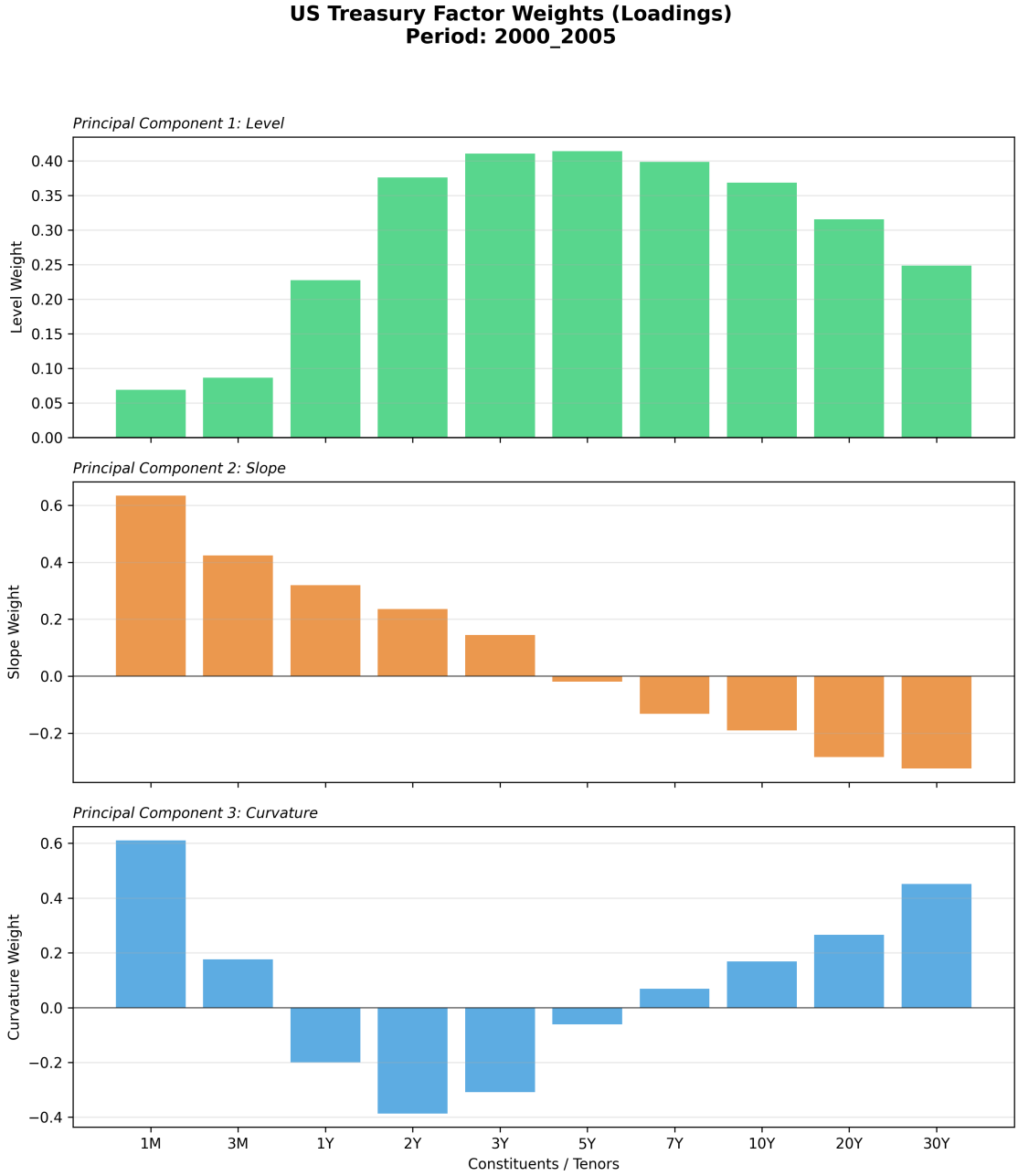


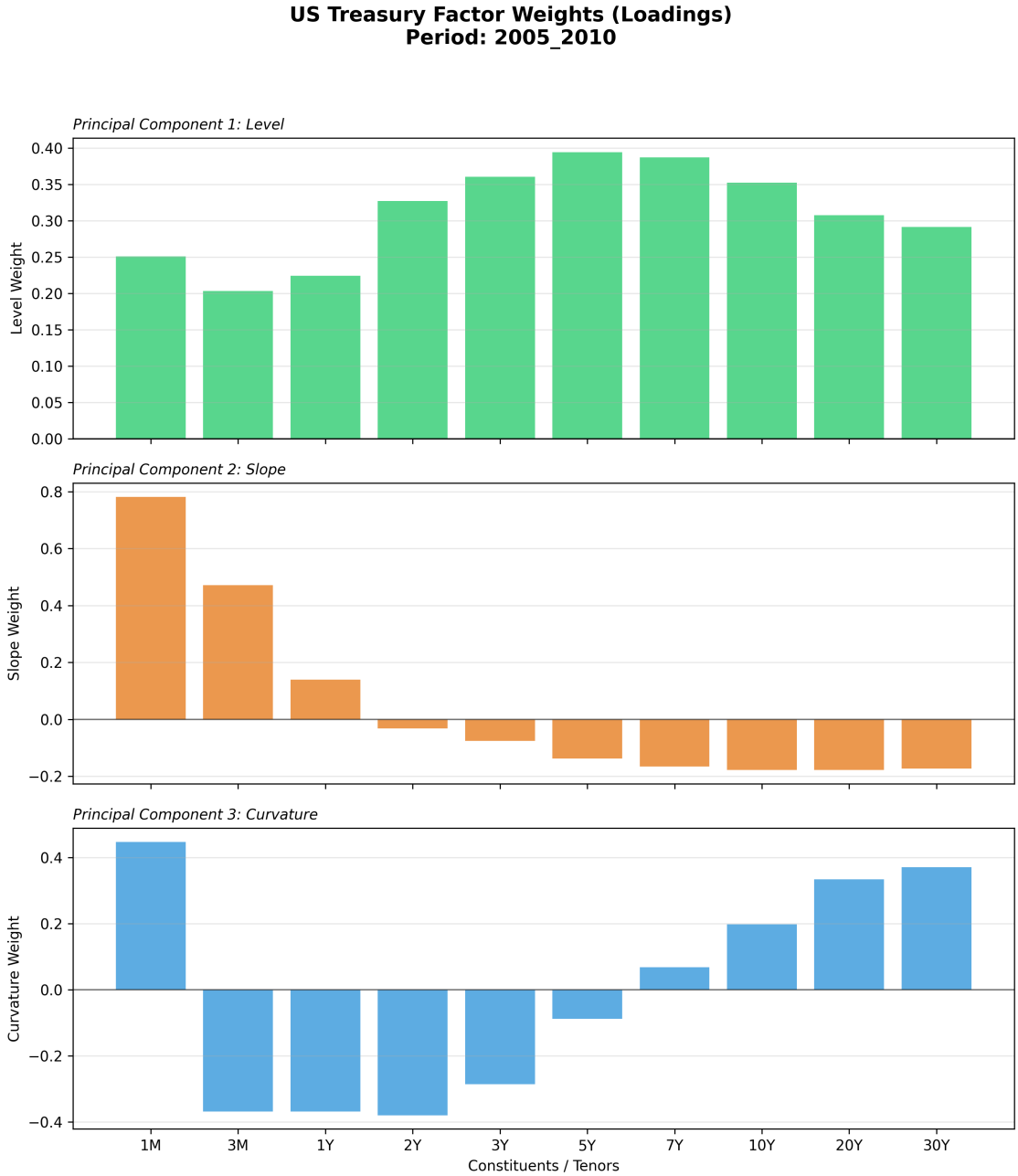


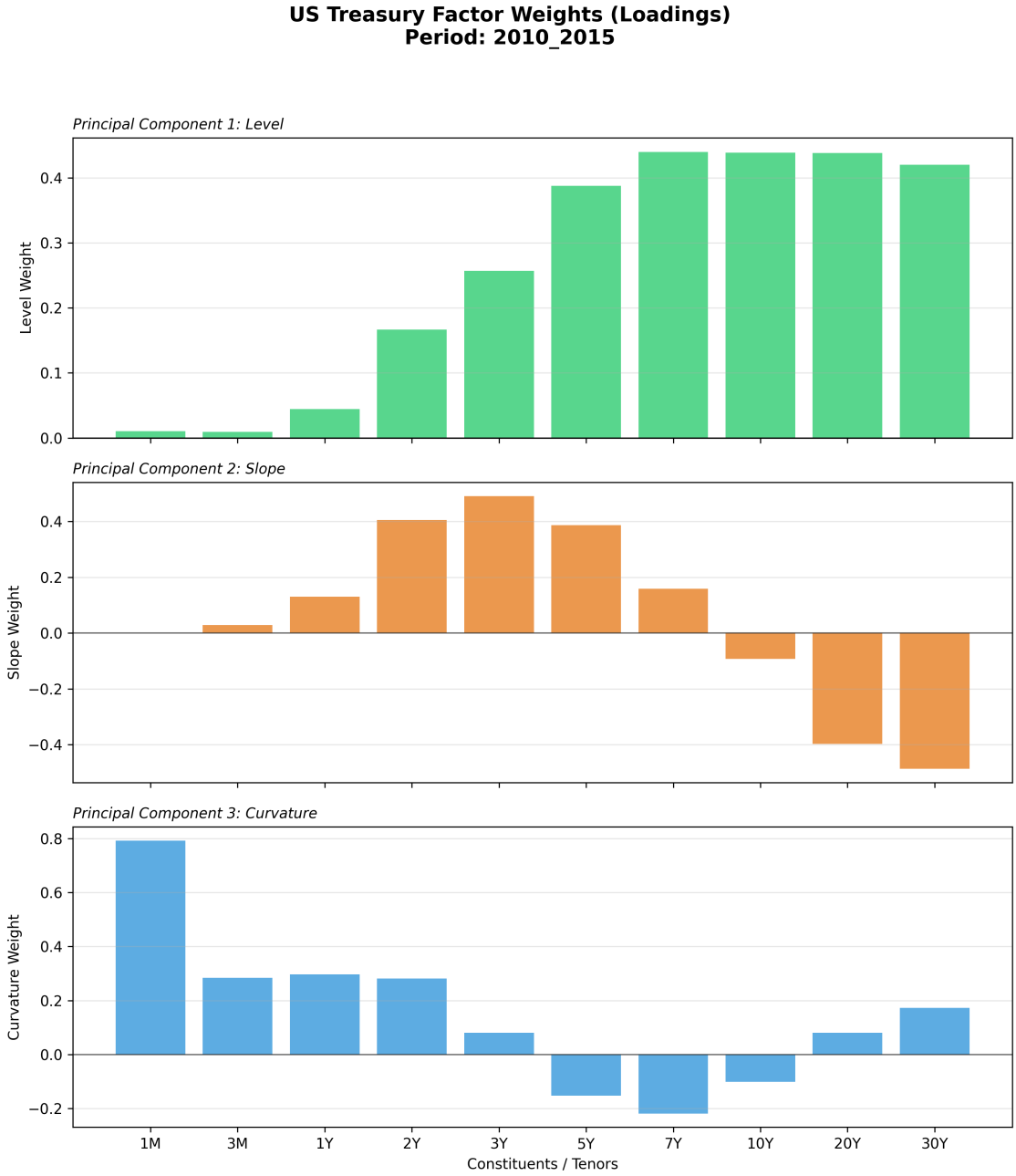

US Treasury Scree Plot: Explained Variance (2015_2020)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
78.3%
8.7%
6.5%
2.9%
1.5%
0.9%
0.4%
0.3%
0.3%
0.2%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10

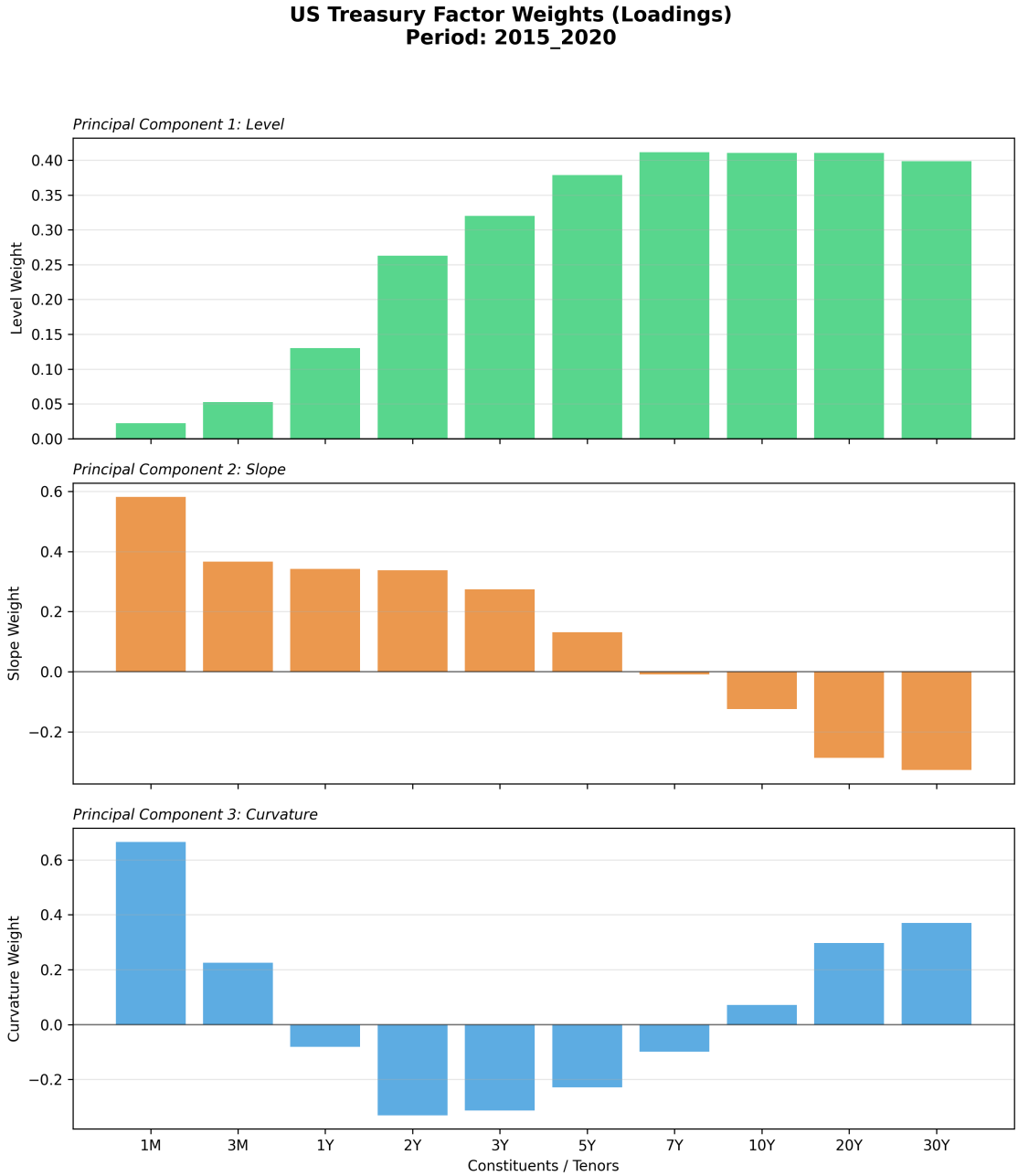

US Treasury Scree Plot: Explained Variance (2020_2025)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
73.2%
10.6%
10.0%
3.4%
1.4%
0.7%
0.3%
0.2%
0.2%
0.1%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10

B Interest Rate: Factor Loadings (2000–2025)




US Treasury Factor Weights (Loadings)
Period: 2005_2010
Principal Component 1: Level
Level Weight
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Principal Component 2: Slope
Slope Weight
0.8
0.6
0.4
0.2
0.0
−0.2
Principal Component 3: Curvature
Curvature Weight
0.4
0.2
0.0
−0.2
−0.4
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors


US Treasury Factor Weights (Loadings)
Period: 2010_2015
Principal Component 1: Level
Level Weight
0.0
0.1
0.2
0.3
0.4
Principal Component 2: Slope
Slope Weight
0.4
0.2
0.0
−0.2
−0.4
Principal Component 3: Curvature
Curvature Weight
0.8
0.6
0.4
0.2
0.0
−0.2
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors


US Treasury Factor Weights (Loadings)
Period: 2015_2020
Principal Component 1: Level
Level Weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Principal Component 2: Slope
Slope Weight
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Curvature
Curvature Weight
−0.2
0.0
0.2
0.4
0.6
1M
3M
1Y
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10Y
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30Y
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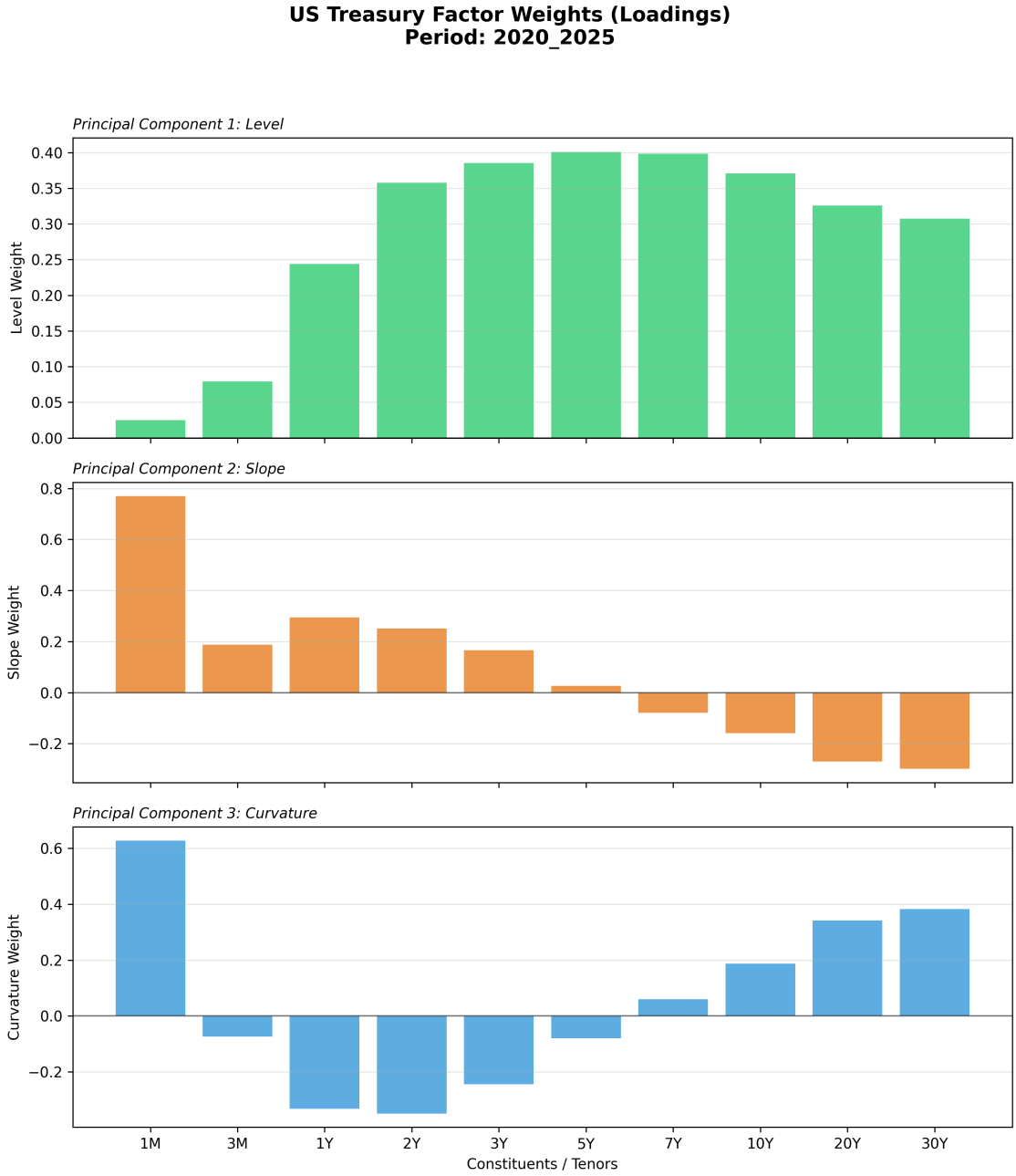
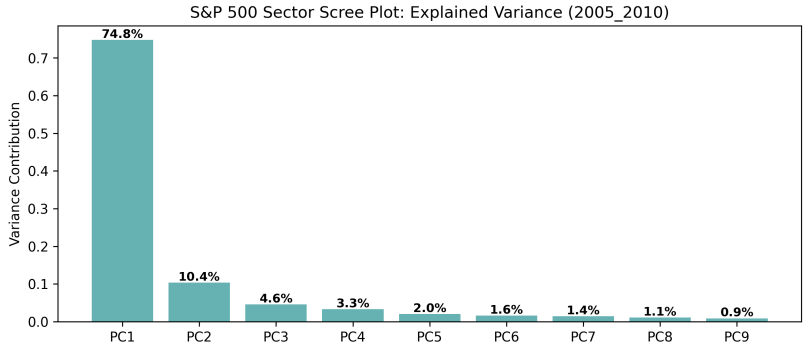
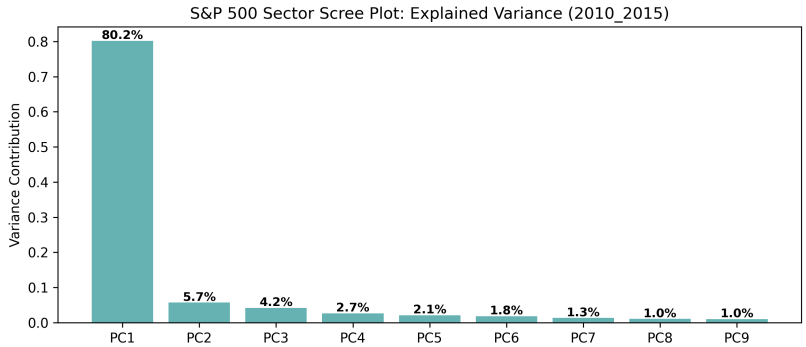
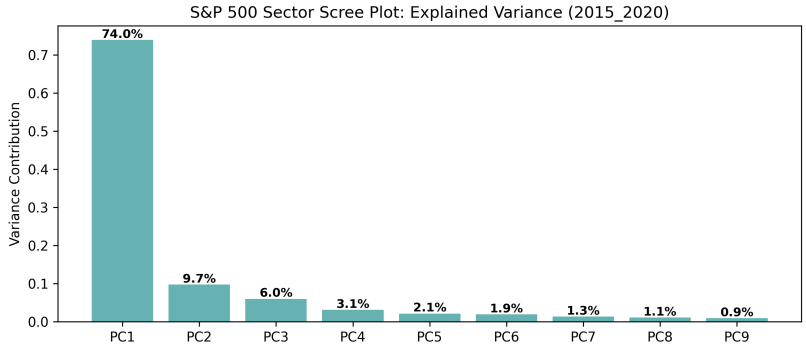
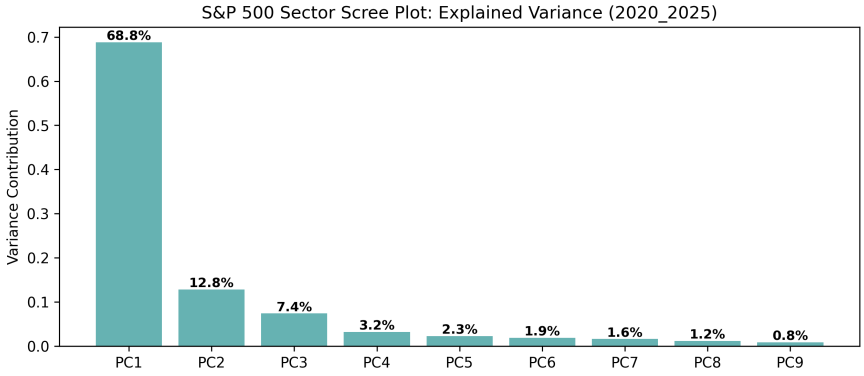

US Treasury Factor Weights (Loadings)
Period: 2020_2025
Principal Component 1: Level
Level Weight
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Principal Component 2: Slope
Slope Weight
0.8
0.6
0.4
0.2
0.0
−0.2
Principal Component 3: Curvature
Curvature Weight
0.6
0.4
0.2
0.0
−0.2
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors

C Equity Sector: Scree Plots (2005–2025)

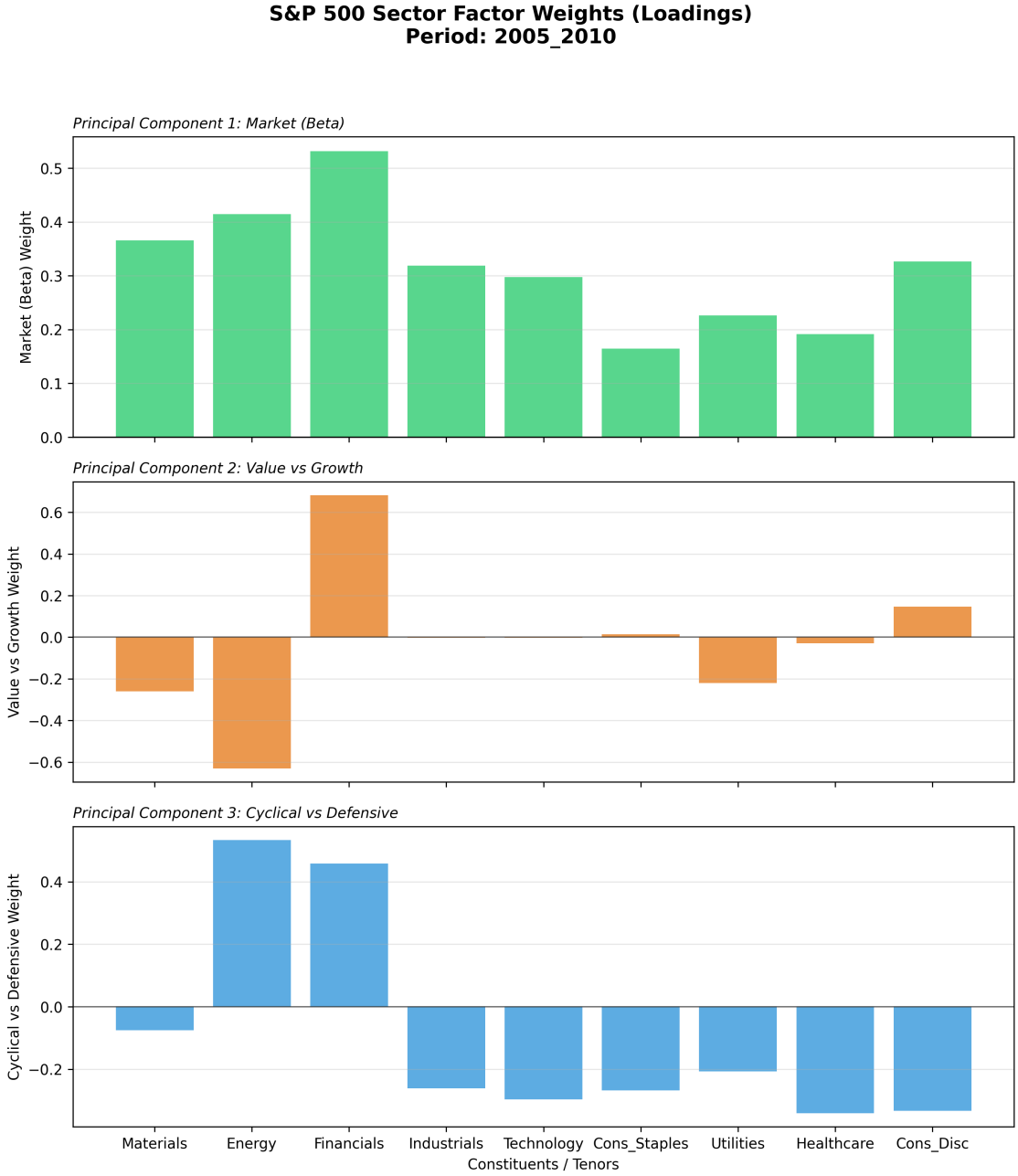




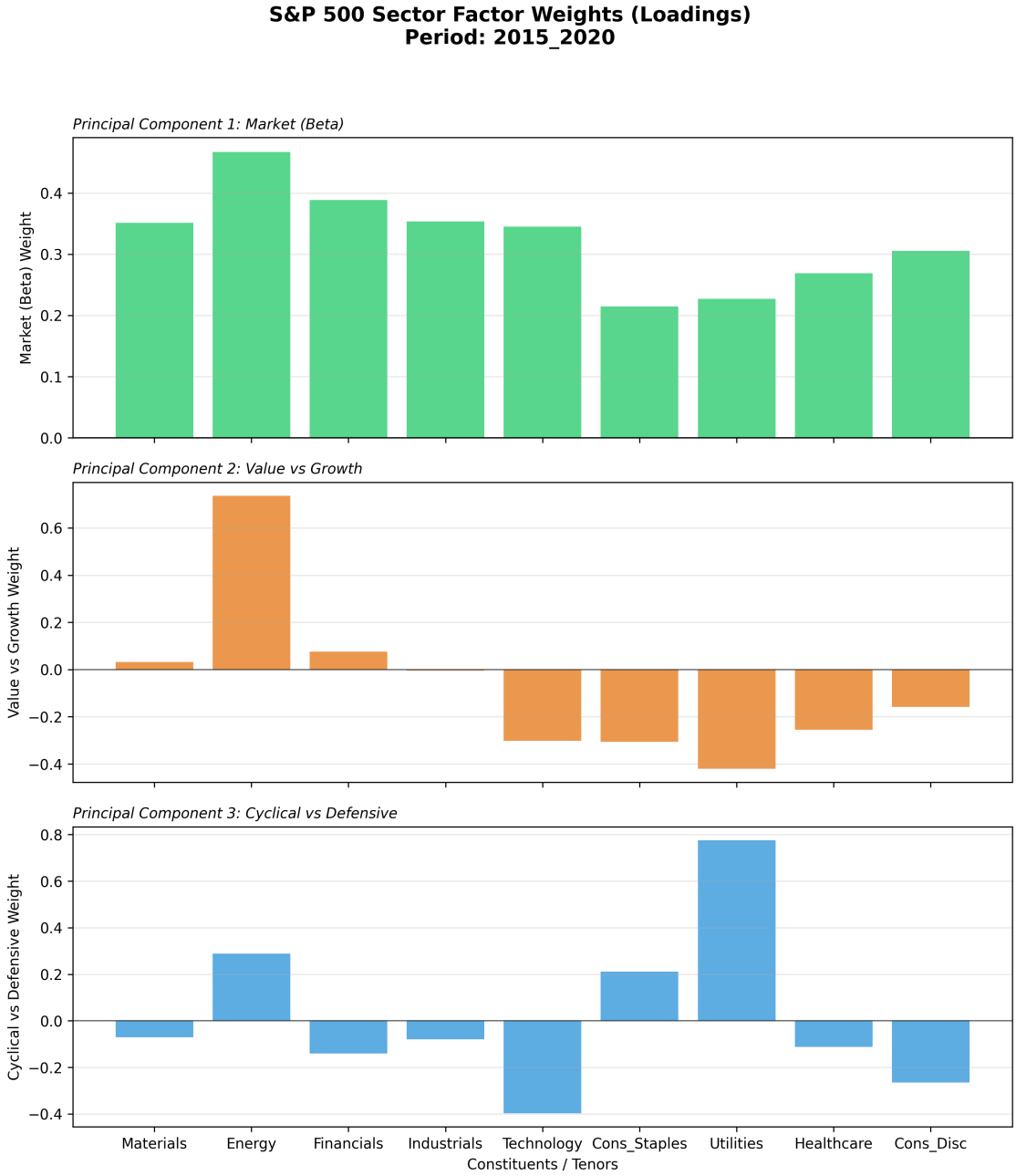



S&P 500 Sector Scree Plot: Explained Variance (2020_2025)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
68.8%
12.8%
7.4%
3.2%
2.3%
1.9%
1.6%
1.2%
0.8%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9

D Equity Sector: Factor Loadings (2005–2025)



S&P 500 Sector Factor Weights (Loadings)
Period: 2010_2015
Principal Component 1: Market (Beta)
Market (Beta) Weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Principal Component 2: Value vs Growth
Value vs Growth Weight
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
−0.2
0.0
0.2
0.4
0.6
0.8
Materials
Energy
Financials
Industrials
Technology
Cons_Staples
Utilities
Healthcare
Cons_Disc
Constituents / Tenors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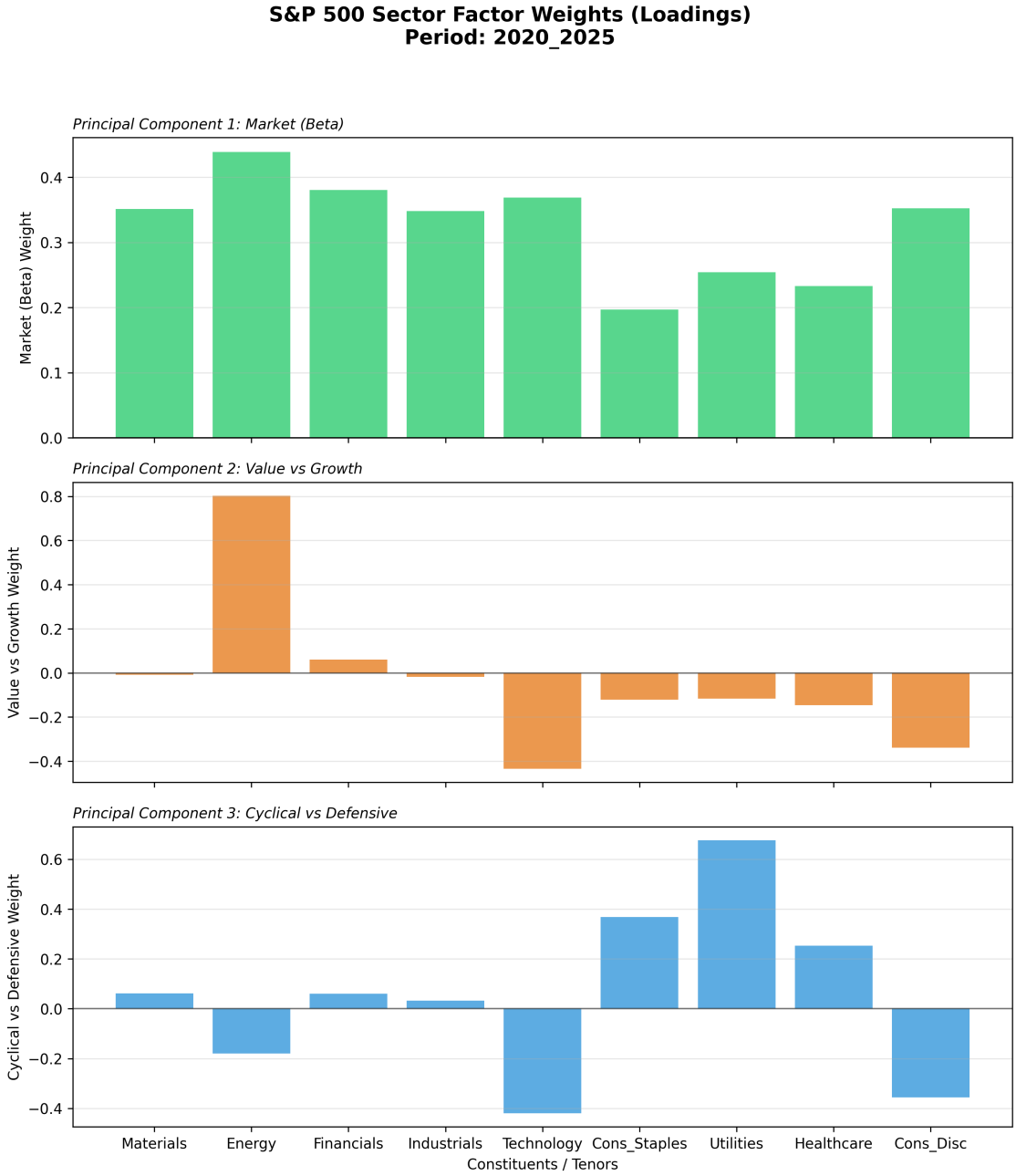
Principal Component 1: Market (Beta)
Market (Beta) Weight
0.0
0.1
0.2
0.3
0.4
Principal Component 2: Value vs Growth
Value vs Growth Weight
−0.4
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
Materials
Energy
Financials
Industrials
Technology
Cons_Staples
Utilities
Healthcare
Cons_Disc
Constituents / Tenors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Period: 2020_2025
Principal Component 1: Market (Beta)
Market (Beta) Weight
0.0
0.1
0.2
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0.4
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Value vs Growth Weight
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
0.6
0.4
0.2
0.0
−0.2
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Materials
Energy
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Technology
Cons_Staples
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Constituents / Tenors